

54.3 - Homework

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Homework 2.3

3. Find inverse using few calculations as possible for

$$\begin{bmatrix} 5 & 0 & 0 \\ -3 & -7 & 0 \\ 8 & 5 & -1 \end{bmatrix}$$

- I fully solved to get identity matrix originally. The intuition behind not solving all the way is to see that any row scaling and adding will not effect other a_{ij} positions within the matrix
- We can also just not reduce them fully (i.e. divide so we get the leading entry to be 1 for the pivot). Thus the answer should look like this:

$$\begin{bmatrix} 5 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

11. If the equation $Ax = 0$ has only the trivial solution, then A is row equivalent to the $n \times n$ identity matrix.

- True, because if it only has trivial solution $\rightarrow$ it has n pivot positions $\rightarrow$ it is row equivalent to $n \times n$ identity matrix.

- $d \implies c \implies b$

13. If the columns of A span $\mathbb{R}^n$ then the columns are linearly independent.

- I originally thought that we had to prove the textbook IMT statements $g \iff h \iff i$, but rather if you can show one statement is true, then it applies for all other ones!
- Recall that if columns of A span $\mathbb{R}^n \rightarrow \forall b \in \mathbb{R}^n, Ax = b$ has a solution.

23. Can a square matrix with two identical columns be invertible, Why or why not?
- By definition, if two columns are identical then it has dependence and thus it will not be invertible.
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Homework 3.1

5. Find determinant of

$$\begin{bmatrix} 2 & 3 & -3 \\ 4 & 0 & 3 \\ 6 & 1 & 5 \end{bmatrix}$$

- Apply the algorithm:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

- You must solve determinant and scale by the constant.
 - Recall that we always define the dimension of $n \times n$ by its inductive definition (i.e. if we had 4×4 , it would be 3×3 that we write it as.
 - answer is 22.
11. Compute the determinants in Exercises 9-14 by cofactor expansions. At each step, choose a row or column that involves the least amount of computation
- Essentially you want to apply the minor algorithm and cofactor expansion (ie. remove columns and rows) to simplify computation.
 - [Great Video](#)
 - **NOTE:** In this example its easy because we dont have to worry about inverting because of swapping rows, or multiplying by a scalar because we scaled a row.
27. Compute the determinants of the elementary matrices given. See examples 5 and 6 from Ch 2.2.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ k & 0 & 1 \end{bmatrix}$$

- This is asking you to perform the elementary matrix (i.e. one operation that makes it I), and thus by doing so we show that its another way to prove $A^{-1}E_k \dots E_1$

- More importantly, **And this is the answer**, is that by theorem 2 if its a triangular matrix, then $\det A$ is the product of entries on the main diagonal of A . Thus the answer is 1.
- **Concept Check**, if one column wasnt present, i.e. it is 0, what happens to the equation?
 - We would apply triangular-matrix theorem and find that $1 \cdot 0 \cdot 1 = 0$. Because column without a pivot means $\exists 0$ on the diagonal that gives determinant zero.

29. Compute the determinants of the elementary matrices given.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- I predict same conclusion in 27, because it is a triangular-matrix theorem. Thus the answer is $1 \cdot k \cdot 1 = k$

31. What is the determinant of an elementary row replacement matrix?

- With question 29 under our belts, we understand that the determinant is:

$$\det(A) = \prod_{i=1}^n a_{ii}$$

- I.e. If A is triangular and has no dependence, you must **multiply all diagonal entries. No exceptions.**

Homework 3.2

3. State the property of the determinant

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & 3 & -4 \\ 2 & 7 & 4 \end{bmatrix}$$

- By performing one row operation to get rid of 2 in row 3, we can see that the diagonal matrix is broken
- Thus, we have shown thats not invertible because $\det(A) = 0$
- We are tested on this because I mixed up subjects. I.e.. In the begining of the chapter we state that **the row replacement operation does not change the determinant (Theorem 3a.)**
- **I didn't show the $\det(A) = 0$, and thus I had a false conclusion**

7. Find determinants with row reduction to echelon form

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ -2 & -5 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{bmatrix}$$

- When we start to put it in echelon form, we see that there is dependence in $R3 \wedge R4$.
- Rows of A are columns of A^T . Thus if we see dependence in rows OR columns then the condition $\det(A^T) \neq 0 \wedge \det(A) \neq 0$ is not satisfied for both.

11. Use row reduction and cofactor expansion to find determinants

$$\begin{vmatrix} 3 & 4 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ -6 & 0 & -4 & 3 \\ 6 & 8 & -4 & -1 \end{vmatrix}$$

- I got 11 wrong because I wasn't understanding this algorithm. When we conduct the row reduction then we must take the entry and scale it by what we find when we apply cofactor expansion.

12. (Extra Practice for 11) Use row reduction and cofactor expansion to find determinants

$$\begin{vmatrix} -1 & 2 & 3 & 0 \\ 3 & 4 & 3 & 0 \\ 11 & 4 & 6 & 6 \\ 4 & 2 & 4 & 3 \end{vmatrix}$$

- Choose the column with most zeros, then apply same idea of finding zeros.
- You should get a two by two matrix that allows you to produce 6 as the answer. I.e.:

$$3 \begin{vmatrix} 10 & 12 \\ 6 & 7 \end{vmatrix}$$

15. - 20. These questions are testing your understanding of what it means to scale, interchange, and add a linear equation with another. The rules to solve the puzzles are:

- If we multiply, we must scale the det found by that scalar.
- If we interchange, we must multiply by a -1.
- If we add, the det stays the same!

25. Use determinants to decide if the set of vectors is linearly dependent

$$\begin{bmatrix} 7 \\ -4 \\ -6 \end{bmatrix}, \begin{bmatrix} -8 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 7 \\ 0 \\ -5 \end{bmatrix}$$

- I got this correct, but the wrong det solution. We know its independent.
- I learned a valuable lesson that minors are a side effect of cofactor expansion. The algorithm must be following explicitly. I.e.:
- If using column, then $\det(A) = \sum_{j=1}^n a_{ij}M_{ij}$
- Applied here:

$$7(-1)^{1+1} \begin{vmatrix} 5 & 0 \\ 7 & -5 \end{vmatrix} \dots + a_i(-1)^{(i+j)}|M_{ij}|$$

28. If $\det(A)$ is zero, then two rows or two columns are the same or a row or a column is zero.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \rightarrow ac - bc = 0 \implies \text{linear dependence}$$

Thus we can see that the columns are not identical or 0 and still produce 0. Making this statement False.